

Impedance Model Predictive Control for Steerable Catheter Tip Control Under Unknown Contact Forces with Hard Safety Constraints

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Abstract—Steerable catheters for cardiac electrophysiology must simultaneously track a planned tip trajectory and maintain safe, compliant contact with tissue—objectives that classical impedance control cannot meet without steady-state error under persistent contact force and with no formal mechanism for enforcing a contact-force safety bound. We present an Impedance Model Predictive Control (Impedance MPC) framework for catheter tip control. Recognizing that a single-segment single-tendon catheter is a single-degree-of-freedom system, the controller is developed in the scalar tip-normal coordinate: a partial-physics feedforward cancels the *known* nominal bending dynamics, reducing the uncertain nonlinear plant to a constant-coefficient double integrator, and a receding-horizon quadratic program (QP) regulates the residual subject to a hard contact-force bound, tendon-force limits, and curvature limits. An augmented Kalman filter lumps tissue contact force, friction, hysteresis, and modeling error into a single estimated disturbance state, rendering tracking offset-free in the nominal limit without a force sensor. We establish equivalence to classical catheter impedance control in the unconstrained, disturbance-free limit (Theorem 1, with proof sketch), and state input-to-state stability under bounded disturbance (Theorems 2–3, whose proofs follow standard offset-free/LPV-MPC arguments). A single-DOF simulation shows that the Kalman augmentation rejects a constant friction/hysteresis bias—cutting free-space approach error by 98% relative to classical impedance (0.50→0.01 mm)—and that, decisively, only the force-constrained MPC keeps the contact force within the 0.5 N safety bound (peak 0.25 N versus 0.79 N for classical impedance and 7.25 N for an unconstrained offset-free controller that drives hard into tissue). We further show that offset-free disturbance rejection and contact-force safety are in direct tension under stiff contact, resolved only by the explicit hard constraint.

Index Terms—Steerable catheter, continuum robot, impedance control, model predictive control, Kalman filter, contact-force safety, offset-free tracking, cardiac ablation.

I. Introduction

A. Clinical Motivation

Steerable catheters are the primary tool for cardiac electrophysiology (EP) procedures including radiofrequency ablation, where the tip must be positioned precisely at target tissue while maintaining controlled, stable contact. Two competing objectives define the control problem: (i) **trajectory tracking**—the tip must reach and hold a planned position with sub-millimeter accuracy so that ablation lesions are placed correctly; and (ii) **safe compliance**—the tip must not exert excessive force against the endocardium, since over-contact risks perforation (a potentially fatal complication [12]) while under-contact yields inadequate lesions.

Classical impedance control [1] shapes the tip port as a mechanical impedance $Z(s) = M_d s^2 + D_d s + K_d$, providing passive compliance without an explicit contact model. It has been applied to catheter tip-force regulation for its simplicity and inherent passivity, but suffers three limitations in this setting: (i) the contact-force safety bound $\|F_{\text{tip}}\| \leq F_{\text{safe}}$ cannot be enforced (post-hoc force clipping destroys passivity); (ii) any persistent contact force produces a steady-state tip error $e_\infty = K_d^{-1} F_{\text{contact}}$, preventing accurate lesion placement; and (iii) the controller cannot anticipate trajectory curvature or approaching contact.

B. Challenges Specific to Catheters

Catheters present challenges absent in rigid-body robots. **Model uncertainty:** the mechanics follow a variant of the Cosserat rod equations [3], [4], with distributed elasticity, sheath friction, tendon backlash, and hysteresis that vary widely across specimens, making full-model inversion impractical.

Cardiac motion: the heart moves ~10–15 mm per cycle at ~1 Hz, so contact forces are quasi-periodic with known frequency but unknown phase and amplitude. **Limited sensing:** most clinical catheters carry no force sensor; tip position is available via electromagnetic (EM) tracking, fluoroscopy, or intracardiac echocardiography at moderate latency (10–50 ms). **Safety criticality:** a perforation force threshold of ~0.3–0.5 N is clinically relevant [12], making hard force-constraint enforcement—not soft penalization—essential.

C. Related Work

Catheter and continuum-robot control has progressed from PID and Jacobian-based kinematic control [5], [6] to model-less feedback [6] and MPC formulations. Cosserat-based MPC achieves high tip accuracy but requires online nonlinear-program solution, limiting update rates to ~10–20 Hz. Constant-curvature kinematics [2] underlie most reduced-order controllers. Learning-based approaches improve robustness to model uncertainty but lack formal stability and constraint guarantees.

Sensor-free contact-force control is an active and directly relevant line. Jolaei *et al.* [14] regulate the contact force of a tendon-driven ablation catheter *without a force sensor* by combining position control with a displacement-based viscoelastic contact model, reporting force-regulation RMS error of 0.03–0.05 N. Kesner and Howe [15] combine ultrasound guidance with force control on a moving cardiac target, achieving ~0.08 N RMS force tracking. These works regulate force to a *setpoint*; the present framework instead embeds force as a *hard never-exceed safety constraint* inside an offset-free position-tracking MPC, a complementary objective (Section VI-C). Note that [15] utilized ultrasound guidance to actively track the moving target, which explains their low force RMS.

Impedance control for catheters has been demonstrated for tip-force regulation with empirically tuned parameters; the configuration-dependent effective impedance is typically not addressed, giving inconsistent compliance across the workspace.

Disturbance rejection in flexible robots. Active disturbance rejection control (ADRC) and its extended state observer [8] lump unknown dynamics into an estimated disturbance state. The connection to offset-free MPC [9], [10] provides a formal framework for combining disturbance estimation with constraint-aware optimization [7].

Base framework. This work builds directly on the constant- A_d offset-free Impedance MPC framework introduced for redundant-manipulator physical human–robot interaction [13], which established the two-layer feedforward-plus-receding-horizon-QP architecture, the augmented-Kalman offset-free guarantee, and the equivalence to classical impedance control. We carry that framework to the steerable-catheter setting, where the contributions below address the catheter-specific challenges of single-DOF actuation, severe model uncertainty, and hard contact-force safety.

D. Contributions

1. **Correct single-DOF formulation.** A single-segment single-tendon catheter has one controllable degree of freedom (the curvature). We develop the controller in the scalar tip-normal coordinate, avoiding the common over-parameterization that treats the 2-D tip as independently actuable.

2. **Partial-physics feedforward.** A feedforward layer cancels only the *known* nominal bending stiffness and damping, leaving a constant-coefficient double integrator. Unlike full computed-torque inversion, this is robust to the large modeling uncertainty of catheter mechanics; the residual error is absorbed by the disturbance state.
3. **Impedance MPC unification.** We establish (Theorem 1) that Impedance MPC recovers the classical catheter impedance law in the unconstrained, disturbance-free limit, while adding constraint enforcement and offset-free disturbance rejection.
4. **Hard contact-force safety constraint.** The QP enforces $\|F_{\text{tip}}\| \leq F_{\text{safe}}$ as a hard bound over the horizon, realized through the transmitted tendon force—a capability unavailable to any impedance controller.
5. **Honest characterization of MPC value.** We show, by simulation, that the dominant benefit of MPC here is *constraint enforcement*, that offset-free rejection helps mainly in free space, and that offset-free tracking *without* a force constraint is actively dangerous under stiff contact.

II. System Model and Problem Formulation

A. Catheter Mechanics and Degrees of Freedom

Consider a planar single-segment tendon-actuated catheter; tip position $p \in \mathbb{R}^2$ is set by tendon displacement $u \in \mathbb{R}$ (positive pull gives curvature $\kappa > 0$). The dominant bending dynamics take the rigid-body-analogous form

$$M(\kappa)\ddot{\kappa} + C(\kappa, \dot{\kappa})\dot{\kappa} + K(\kappa) = u + d_{\text{cat}}, \quad (1)$$

with effective bending inertia M , damping C , nonlinear (hysteretic) stiffness K , and a lumped term d_{cat} collecting tissue contact, friction, backlash, and hysteresis. The correspondence $\kappa \leftrightarrow q$, $K \leftrightarrow G$ bridges to the constant- A_d offset-free Impedance MPC framework established for redundant manipulators in physical human–robot interaction [13], which forms the methodological base of this work; the present paper specializes it to the single-DOF catheter setting.

Degrees of freedom. With a single tendon, the only controllable coordinate is κ ; the tip pose $p(\kappa)$ traces a one-parameter curve, so the two tip coordinates are *not* independently actuable—only motion along the Jacobian direction J_κ is. We therefore control a scalar coordinate (the curvature, or equivalently the tip displacement y along the contact normal) and recover the 2-D pose through the kinematics of Section II-B.

Remark 1 (Disturbance compression). The full mechanics are an infinite-dimensional Cosserat PDE; (1) is a deliberate reduction in which we control only the tip state $[\kappa, \dot{\kappa}]$ and absorb higher-order distributed effects into d_{cat} . Not every phenomenon must be modeled—only observed and slowly varying relative to the control bandwidth.

B. Tip Kinematics

For a constant-curvature arc of length L ,

$$p_x = \frac{\sin(\kappa L)}{\kappa}, \quad p_z = \frac{1 - \cos(\kappa L)}{\kappa}, \quad (2)$$

with translational Jacobian $J_\kappa(\kappa) = dp/d\kappa$ and operational-space inertia scalar $\Lambda(\kappa) = (J_\kappa M^{-1} J_\kappa^\top)^{-1} > 0$, which normalizes the effective tip impedance to be configuration-adaptive.

C. Contact Model and Safety Constraint

During contact, a tip force F_{tip} acts at the tip; in simulation the tissue is Kelvin–Voigt, $F_{\text{tissue}} = k_t \delta + b_t \dot{\delta}$, with penetration $\delta = \max(0, p_{\text{surf}} - p)$, tissue stiffness $k_t \approx 2\text{--}20$ kN/m, and damping b_t . The contact maps to curvature via $\tau_{\text{ext}} = J_\kappa^\top F_{\text{tip}}$. The clinically relevant perforation threshold $F_{\text{safe}} \approx 0.5$ N [12] must be a hard bound, $\|F_{\text{tip}}(t)\| \leq F_{\text{safe}} \forall t$.

D. Control Objective

Given a reference $(p_d, \dot{p}_d, \ddot{p}_d)$, design $u(t)$ so the tracking error $e = p_d - p$ (projected onto the controllable direction) satisfies $e(t) \rightarrow 0$, subject to tendon limits $u_{\text{lo}} \leq u \leq u_{\text{hi}}$, force safety $\|F_{\text{tip}}\| \leq F_{\text{safe}}$, and curvature limits $\kappa_{\text{lo}} \leq \kappa \leq \kappa_{\text{max}}$, under bounded unknown contact, friction, and hysteresis.

III. Impedance MPC Controller Design

A. Two-Layer Architecture

$$u = \underbrace{u_{\text{ff}}}_{\text{Layer 1: partial-physics cancellation}} + \underbrace{J_\kappa^\top F_{\text{mpc}}}_{\text{Layer 2: MPC correction}}, \quad (3)$$

with feedforward $u_{\text{ff}} = \hat{C}(\kappa, \dot{\kappa})\dot{\kappa} + \hat{K}(\kappa) + J_\kappa^\top \Lambda(\kappa)\ddot{p}_d$ canceling the *nominal* dynamics. In the single-DOF setting $F_{\text{mpc}} \in \mathbb{R}$ is the scalar corrective force along the controlled tip-normal, and

$J_\kappa \equiv dp_y/d\kappa \in \mathbb{R}$ is the corresponding scalar Jacobian component (tip-normal displacement per unit curvature); $J_\kappa^\top F_{\text{mpc}}$ is thus a scalar product mapping the tip-normal force to a curvature-space (tendon) moment, dimensionally consistent with u . For a rigid robot Layer 1 is exact; for a catheter $\hat{C}$, $\hat{K}$ are uncertain, so the cancellation is partial and the residual is absorbed by the disturbance estimate—more robust than inverting an unreliable model.

B. Scalar Error Dynamics and LPV Model

Since the system is single-DOF (Section II-A), the error $e = y_d - y$ and the corrective input $F_{\text{mpc}} \in \mathbb{R}$ are scalar. With $x_e = [e, \dot{e}]^\top \in \mathbb{R}^2$,

$$\dot{x}_e = \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}}_{A_c \text{ (const)}} x_e + \underbrace{\begin{bmatrix} 0 \\ -\Lambda^{-1}(\kappa) \end{bmatrix}}_{B_c(\kappa)} F_{\text{mpc}} + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{E_c \text{ (const)}} d(t), \quad (4)$$

where d lumps modeling error, contact, and the kinematic coupling $\dot{J}_\kappa \dot{\kappa}$, normalized to acceleration units. A_c is **constant**; only the scalar gain $B_c(\kappa)$ varies through $\Lambda(\kappa)$ —the same constant- A_d , parameter-varying- B_d structure exploited in the base framework [13] and, earlier, in min-max MPC under input saturation [11], enabling offline precomputation of the prediction matrices. Because A_c is nilpotent, the exact ZOH discretization is closed-form:

$$A_d = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}, \quad B_d(\kappa_k) = \begin{bmatrix} -\frac{1}{2}\Lambda^{-1}(\kappa_k)\Delta t^2 \\ -\Lambda^{-1}(\kappa_k)\Delta t \end{bmatrix}. \quad (5)$$

C. Disturbance Augmentation

Augment with a scalar integrating disturbance $\hat{d} \in \mathbb{R}$:

$$\begin{bmatrix} x_e(k+1) \\ \hat{d}(k+1) \end{bmatrix} = \begin{bmatrix} A_d & G_d \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_e(k) \\ \hat{d}(k) \end{bmatrix} + \begin{bmatrix} B_d(\kappa_k) \\ 0 \end{bmatrix} F_{\text{mpc}}(k), \quad (6)$$

with $G_d = [\frac{1}{2}\Delta t^2, \Delta t]^\top$ the ZOH of E_c —**distinct from** B_d , which carries the $-\Lambda^{-1}$ input gain (d is in acceleration units, so G_d is not Λ^{-1} -scaled). A steady-state Kalman filter estimates $\hat{d}$ from tip-error measurements; the pair is observable with $C_{\text{aug}} = [I_2, 0]$ since $G_d \neq 0$. No force sensor is required.

D. Receding-Horizon QP

$$\min_U \frac{1}{2} U^\top H U + (\Phi x_e(k) + \Delta(\hat{d}))^\top \bar{Q} \Gamma U, \quad (7)$$

with $U = [F_{\text{mpc}}(0); \dots; F_{\text{mpc}}(N-1)] \in \mathbb{R}^N$, $H = \Gamma^\top \bar{Q} \Gamma + \bar{R}$, $Q = \text{diag}(K_d, D_d)$, and $\Phi \in \mathbb{R}^{2N \times 2}$, Γ precomputed once (constant A_d). When no constraint is active the solution is the closed form $U^* = -H^{-1} \Gamma^\top \bar{Q} (\Phi x_e + \Delta(\hat{d}))$ —a matrix–vector multiply; when a constraint binds, OSQP solves the condensed QP reusing the cached factorization. Subject to tendon limits, curvature limits, and the contact-force bound.

Force-safety constraint. At contact equilibrium the feedforward cancels nominal stiffness/damping, so $F_{\text{tip}} = u - \hat{F}_{\text{fric}}$, where $\hat{F}_{\text{fric}} = \Lambda \hat{d}_{\text{free}}$ is the calibrated free-space friction bias (separated from contact by the Kalman estimate). The hard bound becomes a linear cap on the command, $u_{\text{ff}}(k) + J_\kappa^\top F_{\text{mpc}}(k) \leq F_{\text{safe}} + \hat{F}_{\text{fric}}$.

E. Impedance MPC Equivalence

Theorem 1. Let $Q = \text{diag}(K_d, D_d)$ and $R \rightarrow 0$. With no constraints and $\bar{d} \equiv 0$, the receding-horizon solution as $N \rightarrow \infty$ converges to the classical catheter impedance law $u_{\text{imp}} = \hat{C} \dot{\kappa} + \hat{K} + J_\kappa^\top (K_d e + D_d \dot{e})$, with effective tip impedance $Z_{\text{eff}}(s, \kappa) \approx \Lambda(\kappa)(s^2 + D_{\text{eff}} s + K_{\text{eff}})$.

Proof sketch. Without constraints and $R \rightarrow 0$, the QP reduces to unconstrained minimization of $\sum_k x_e^\top Q x_e$; the infinite-horizon optimum is the LQR gain $[K_d, D_d]$ for the double integrator, recovering $F_{\text{mpc}} = K_d e + D_d \dot{e}$ at the nominal configuration. $\square$

When constraints are active or $\bar{d} \neq 0$, the MPC departs from the impedance law—precisely its advantage.

IV. Stability Analysis

Theorem 2 (Nominal stability). With terminal cost Q_f the DARE solution at a nominal configuration κ_0 , if the QP is feasible at $k = 0$ and the LPV variation $\|B_d(\kappa_k) - B_d(\kappa_0)\|$ is sufficiently small, the closed loop is asymptotically stable and $x_e(k) \rightarrow 0$ (no disturbance), provided recursive feasibility holds.

Theorem 3 (ISS / offset-free). If $\|d(t)\| \leq \bar{d}$ and the augmented system is observable, $(x_e, \hat{d})$ is input-to-state stable and $\|e_\infty\| \rightarrow 0$ as the Kalman filter converges—in the nominal, free-space limit. Under stiff contact the achievable steady-state error is additionally bounded by the force constraint and is therefore nonzero (Section VI-B confirms a contact-limited residual).

Proof status. Theorem 1 is supported by the sketch above. Theorems 2–3 follow standard arguments—terminal-cost/recursive-feasibility stability for constrained MPC [7] and the augmented-observer

offset-free result [9]—specialized to the constant- A_d scalar plant.

Remark 2 (Quantified LPV margin). The "sufficiently small variation" condition of Theorem 2 admits a concrete bound for this scalar plant. Since $B_d(\kappa) \propto \Lambda^{-1}(\kappa)$, a gain K designed at Λ_{ref} applied where the true inertia is Λ_{true} yields the closed loop $A_d - \rho B_d(\Lambda_{\text{ref}})K$ with $\rho = \Lambda_{\text{ref}}/\Lambda_{\text{true}}$. Computing the spectral radius (`catheter_verify.py`), the loop remains Schur for $\rho < 3.4$, i.e. for $\Lambda_{\text{true}} > 0.29 \Lambda_{\text{ref}}$; under-estimates ($\Lambda_{\text{true}} > \Lambda_{\text{ref}}$) are always stable. Over the workspace $\Lambda(\kappa)$ varies only $1.4\times$ ($\rho \in [0.70, 1.0]$)—comfortably inside the stable region—so a single fixed gain is robustly stable, and the $\Lambda(\kappa)$ -normalized gain is configuration-invariant to machine precision (Section VI-F). The nominal-limit caveat on Theorem 3 is essential: as the simulation shows (Sections VI-B, VI-E), offset-free tracking does **not** hold under stiff contact or fast cardiac motion.

V. Discussion

A. The Role of MPC Prediction

Prediction source	Quality	Benefit
Force prediction $\hat{F}(k+i)$	poor (tissue unknown)	limited
Trajectory feedforward $\ddot{p}_d(k+i)$	exact (offline plan)	high
Constraint anticipation	exact (known bounds)	high
Kalman disturbance estimate at k	moderate	high

The primary value of MPC here is **constraint enforcement** and **trajectory feedforward**, not contact-force prediction; a zero-order-hold $\hat{d}(k+i) = \hat{d}(k)$ is adopted, and the offset-free guarantee is preserved at steady state regardless of prediction quality.

B. The $\dot{d} = 0$ Assumption and Cardiac Motion

The integrating model is a worst-case approximation guaranteeing offset-free tracking for any slowly varying disturbance. For the ~ 1 Hz quasi-periodic cardiac force, two refinements are possible: (A) increase the disturbance process-noise covariance for faster adaptation (trading noise sensitivity); (B) with ECG-gated phase, a periodic disturbance model $\hat{d}(k+1) = \hat{d}_{\text{DC}}(k) + A_{\text{card}} \sin(\omega_{\text{heart}}(k+1)\Delta t + \phi)$ turns cardiac force into a predictable component—left to future

work. During first contact the estimate lags by several samples; the hard force constraint provides the safety backstop in this transient.

C. Partial vs. Full Physics Cancellation

Catheter stiffness $K(\kappa)$ is subject to hysteresis, temperature dependence, and specimen variability; inverting with an inaccurate $\hat{K}$ can *increase* the effective disturbance. Canceling only reliably estimated components is more robust—the residual simply adds to $d(t)$. Design rule: include a feedforward term only if its omission would push the steady disturbance beyond the Kalman tracking bandwidth.

D. Sensing Requirements

Tip position for Kalman updates is available from EM tracking (10–40 Hz, ~ 1 mm), fluoroscopy (intermittent), ICE (operator-dependent), or FBG shape sensing (continuous, ~ 0.5 mm). A force sensor is *not* required; if FBG force estimation is available, $\hat{F}_{\text{tip}}$ can be injected directly to reduce contact-transient lag.

E. Robustness to Contact-Normal Misalignment

The scalar coordinate y is taken along an assumed contact normal. Proximal deflections along the femoral access path can rotate the true tissue normal by an angle θ relative to this assumed direction. Two effects follow, both benign for small-to-moderate θ . First, the *effective* contact stiffness seen along the assumed axis scales as $k_t \cos^2 \theta$: a 15° misalignment changes it by $\cos^2 15^\circ \approx 0.93$, a 7% error that is exactly the kind of slowly-varying model mismatch the disturbance state $\hat{d}$ absorbs (Remark 1). Second, a tangential component $\sim F \sin \theta$ appears off-axis; for the single-DOF controller this is an unobserved direction, but it does not enter the regulated coordinate and, being bounded, leaves the offset-free and ISS properties intact. For the force-safety bound the projection is *conservative in the wrong direction*: the estimated normal force underestimates the true magnitude by $1/\cos \theta$, so a misalignment-aware design should either tighten F_{safe} by $\cos \theta_{\text{max}}$ or estimate θ from the shape sensor. Quantifying tracking under large, time-varying θ (and the multi-DOF extension that makes the normal observable) is future work.

VI. Simulation Results

A. Setup

We implement the single-DOF tip-normal model in Python (NumPy/SciPy); after partial-physics feedforward the residual plant is the scalar double integrator $\Lambda \ddot{y} = u + F_{\text{ext}} - b_{\text{eff}} \dot{y} - k_{\text{eff}} y$, integrated with RK4. Parameters: $\Lambda = 1.0$, nominal $b_{\text{eff}} = 5$, $k_{\text{eff}} = 50$ (both cancelled by feedforward), constant tendon-hysteresis bias $F_{\text{fric}} = 0.45$ N. Tissue: Kelvin–Voigt wall at 6 mm, $k_t = 5$ kN/m, $b_t = 40$ N·s/m. MPC at 500 Hz, $N = 20$; tendon limit ± 8 N; $F_{\text{safe}} = 0.5$ N. The reference approaches the surface (1 s), presses to a firm-contact target 1.5 mm *past* the surface (≈ 7.5 N if tracked rigidly), holds 2 s (dwell), and retracts. The penetrating target deliberately stresses the safety mechanism. Reproducible: `simulation/catheter_benchmark.py`.

B. Controller Comparison

TABLE I: Four-Controller Simulation Comparison

Controller	Approach RMS (mm)	Max contact force (N)	F_{safe} violated?	Hold pos. error (mm)
Classical impedance	0.50	0.79	Yes	1.35
Impedance MPC (no FC)	0.01	7.25	Yes	0.07
Impedance MPC (with FC)	0.01	0.25	No	1.46
Joint-space PD	0.69	0.55	Yes	1.40

Key results:

- **Offset-free rejection of the friction bias (free space).** Classical impedance leaves $e_{\infty} = F_{\text{fric}}/K_d = 0.45/900 = 0.50$ mm; the offset-free Kalman drives this to 0.01 mm (**a 98% reduction**)—approaching the nominal-limit zero-error guarantee.
- **Force safety is the decisive contribution.** Only the force-constrained MPC respects the 0.5 N bound (peak 0.25 N); classical impedance reaches 0.79 N (58% over).
- **Offset-free *without* a force limit is dangerous.** The unconstrained Impedance MPC, driving hard to eliminate position error against the penetrating target, pushes to 7.25 N—over 14× the bound (and reduces its own hold error to 0.07 mm precisely *because* it forces through the

tissue). Offset-free rejection and force safety are in direct tension; only the explicit hard constraint resolves it.

- **Hold error is contact-limited, not controller-limited.** All controllers retain $\sim 1.3\text{--}1.5$ mm hold error because the commanded depth cannot be reached without exceeding F_{safe} ; the force-constrained MPC accepts marginally more (1.46 mm) as the intentional price of safety. The simulation therefore does **not** support a blanket "zero steady-state error" claim under stiff contact—offset-free holds in free space and in the nominal limit.

C. Comparison with Published Catheter-Control Approaches

Table II positions the proposed Impedance MPC against representative classes of catheter/continuum-robot control from the literature. Direct numerical comparison is limited—reported conditions (platform, sensing, contact, metric) differ substantially—so the table compares *capabilities* and order-of-magnitude characteristics rather than asserting head-to-head accuracy. The literature entries (e.g., the " $\sim 10\text{--}20$ Hz" Cosserat-NMPC rate) are **representative characterizations and have not been independently verified** against the primary sources; only the row for this work is from our own simulation (Section VI-B).

TABLE II: Capability Comparison with Representative Catheter-Control Methods

Method (class)	Model	Update rate	Hard force bound	Offset-free	Force sensor	Formal stability
Classical impedance [1]	port impedance	kHz	No	No	optional	passivity
Jacobian / model-less [5], [6]	kinematic	$\sim 10\text{--}100$ Hz	No	No	local	partial
Cosserat NMPC [4]	full Cosserat	$\sim 10\text{--}20$ Hz	soft only	varies	yes	NLP-dependent
Learning-based	data-driven	high	No	No	varies	none
ADRC/ESO [8]	lumped-disturbance	kHz	No	yes	yes (ESO)	yes

Method (class)	Model	Update rate	Hard force bound	Offset- free	Force sensor	Formal stability
Impedance MPC (this work)	reduced + Kalman	500 Hz	Yes (hard)	yes (nominal)	no	ISS + DARE

Observations. (1) Among the surveyed classes, only the proposed framework provides a *hard* contact-force constraint with formal feasibility, the clinically essential capability (Section VI-B). (2) It retains a high update rate (500 Hz) because the constant- A_d structure makes the unconstrained step a matrix–vector multiply, unlike Cosserat NMPC whose online NLP limits rates to $\sim 10\text{--}20$ Hz. (3) It requires no force sensor, unlike ADRC formulations that key off a measured force channel. (4) Its formal guarantees (ISS, DARE-based stability) exceed those of learning-based and purely kinematic controllers. The trade-off is reduced model fidelity relative to full Cosserat MPC—mitigated by the disturbance-compression principle (Remark 1).

Quantitative benchmark. The closest published baselines are the sensor-free tendon-driven ablation-catheter force controllers of Jolaei *et al.* [14] and Kesner and Howe [15]. Table III places the present work against their reported numbers. A direct head-to-head is *not* meaningful because the objectives differ: [14], [15] **regulate** contact force to a setpoint, whereas this work tracks a position trajectory while **bounding** the contact force as a hard constraint. The comparison should therefore be read by capability, not by the force-error column alone.

TABLE III: Quantitative Comparison with Reported Sensor-Free / Force-Controlled Catheter Results

Work	Objective	Sensing	Force result	Position result	Hard force bound
Jolaei <i>et al.</i> [14]	force regulation	sensor-free (contact model)	0.03–0.05 N RMSE	—	no (setpoint)
Kesner & Howe [15]	force regulation	force sensor + ultrasound	~ 0.08 N RMSE	—	no (setpoint)
This work (safety)	position tracking +	sensor-free (Kalman)	bound held, peak 0.25 N ($\leq$	0.01 mm approach	yes (hard)

Work	Objective	Sensing	Force result	Position result	Hard force bound
mode)	force safety		0.5)	RMS	
This work (force-reg mode)	force regulation to 0.2 N + bound	sensor-free (Kalman)	~130 mN RMSE (idealized); ~190 mN (cardiac)	—	yes (hard)

Honest reading. Two points must be stated plainly. (1) The same formulation can also *regulate* force, and the hard safety bound coexists with a regulation setpoint in one controller (Section VI-E); however, with the offset-free position drive now correctly propagated through the horizon, the position-tracking objective and the force-regulation objective compete, and the simple integral force law regulates to only ~130 mN RMSE even in idealized simulation—the force-regulation mode requires a dedicated re-tuning (or an explicit position/force task hierarchy) to recover tight setpoint tracking, which is left as future work. We therefore make **no** claim of force-regulation parity with [14], [15] (whose 30–80 mN are *hardware* results). (2) When 1 Hz cardiac motion is injected—violating the zero-order-hold $\dot{d} = 0$ disturbance model (Section V-B)—force-regulation error is ~190 mN with intermittent contact loss, motivating the cardiac-phase-aware periodic model of Section V-B as essential future work. The distinct, defensible contribution is the unification of offset-free position tracking with a *hard never-exceed* force bound and feasibility guarantees in a single constrained optimization—a capability setpoint force regulation does not provide.

D. Disturbance Estimation

The Kalman filter converges within a few sample periods; during the contact-onset transient the estimate lags, but the tendon-force cap prevents bound violation before convergence—the constraint, not the estimate, provides the safety guarantee. The disturbance estimate improves *tracking*; the hard constraint provides *safety*.

E. Force-Regulation Mode and the Cardiac-Motion Limit

To compare like-for-like with the setpoint force controllers of [14], [15], we add a force-regulation mode: during free-space approach the controller tracks position as before, and during contact it regulates the sensor-free contact-force estimate $\hat{F}_c = \Lambda \hat{d} - \hat{F}_{\text{fric}}$ to a clinical target of 0.2 N via an integral law on the tendon command, while the hard bound $u \leq F_{\text{safe}} + \hat{F}_{\text{fric}}$ remains active. Two conditions are run over the contact-hold window:

- **Idealized (slowly varying disturbance):** force RMSE **~130 mN** about the 0.2 N target. With the disturbance propagation corrected, the offset-free position drive and the force-regulation integral law compete (the controller simultaneously tries to reach the penetrating position target and hold 0.2 N), so the simple integral law no longer regulates tightly; recovering low force RMSE requires re-tuning the force-regulation gain or an explicit position/force task hierarchy (future work). The hard never-exceed bound is unaffected.
- **Realistic (1 Hz cardiac wall motion, 0.3 mm amplitude, plus 0.2 mm position noise):** force RMSE is **~190 mN** with intermittent contact loss. The zero-order-hold $\dot{d} = 0$ model cannot anticipate the periodic motion (Section V-B), so the estimate lags and contact is lost on the retracting half-cycle. This is *worse* than the reported hardware controllers and identifies the cardiac-phase-aware periodic disturbance model (Section V-B, Option B) as **essential** future work for force-regulation use—not merely an enhancement.

The safety-mode results (Table I) are unaffected by this limitation: the hard bound concerns controller-induced force and is enforced regardless of disturbance model, whereas tight force *regulation* under cardiac motion additionally requires predicting the motion.

F. Verification of Structural Claims

Three claims not exercised by the constant- Λ position-tracking benchmark above are verified separately (`simulation/catheter_verify.py`):

1. **Configuration-adaptive compliance.** Computing $\Lambda(\kappa)$ from the constant-curvature kinematics (2) over $\kappa \in [2, 25] \text{ m}^{-1}$ (a $1.4\times$ inertia variation), the $\Lambda(\kappa)$ -normalized controller holds the closed-loop poles **constant to machine precision** (spread $< 2 \times 10^{-6}$), whereas a fixed- Λ gain drifts (and the drift grows with control authority); Fig. (`catheter_lambda_poles.png`) plots both pole loci. This confirms that the $\Lambda(\kappa)$ normalization—not manual gain scheduling—is what renders the tip compliance configuration-independent. The magnitude of the fixed-gain drift is modest for a single segment's $1.4\times$ range but grows for the larger inertia variation of multi-segment catheters (Section VII).
2. **Hard constraint via the QP.** For a 3 mm error the unconstrained step demands $|F_{\text{mpc}}| = 1.99 \text{ N}$; solving the condensed QP with OSQP under a 0.5 N limit returns $|F_{\text{mpc}}| = 0.50 \text{ N}$ (constraint respected), and reproduces the closed-form solution to 10^{-3} when the constraint is inactive. This verifies that the actuator/safety limits are genuinely enforced by the QP, with the closed-form step (Section III-D) recovered when no constraint binds.
3. **Real-time feasibility.** The unconstrained closed-form step runs in $\approx 0.9 \mu\text{s}$ and the warm-started OSQP solve in $\approx 27 \mu\text{s}$ per step on a desktop CPU—both far inside the 2 ms (500 Hz) budget, supporting the real-time claim with margin for embedded hardware. (Embedded-target timing remains future work.)

4. **Approach-velocity safety limit.** A sweep of constant-velocity impacts (FC controller) shows the contact force stays within F_{safe} for approach speeds up to ≈ 7 mm/s (peak 0.49 N at 7 mm/s) and breaches it at 8 mm/s (0.54 N). This is consistent with the damping-only bound $v < F_{\text{safe}}/b_t = 12.5$ mm/s, the simulated limit being lower because penetration adds to the $b_t\dot{y}$ term. The qualification is important for the safety claim: **the hard constraint bounds controller-induced force, but a fast impact injects an environment-induced $b_t\dot{y}$ transient that no input limit can prevent within one sample.** Safe operation therefore requires the approach to be velocity-limited (or the contact onset to be detected and the reference ramped)—a planning constraint, not a controller failure.

VII. Conclusion

We presented an Impedance MPC framework for steerable catheter tip control—specializing the constant- A_d offset-free framework of [13] to the single-DOF catheter setting—developed in the correct scalar coordinate, that extends classical impedance control with hard contact-force constraints, offset-free disturbance rejection (nominal limit), and configuration-adaptive compliance via $\Lambda(\kappa)$. The framework recovers the classical impedance law in the unconstrained limit while adding clinically essential safety. Simulation shows the Kalman augmentation cuts free-space approach error by 98%, and—decisively—that only the force-constrained MPC keeps contact force within the 0.5 N bound, whereas an unconstrained offset-free controller pushes to over 7 N. We characterized this honestly: offset-free disturbance rejection and contact-force safety are in tension under stiff contact, and the hard constraint is what resolves it.

Future work: (i) hardware validation on a tendon-actuated catheter with EM tracking; (ii) FBG-based force estimation as a direct disturbance channel; (iii) cardiac-phase-aware periodic disturbance models; (iv) extension to multi-segment catheters with m independent tendons (an m -DOF version with configuration-varying $\Lambda(\kappa)$); and (v) energy-budget augmentation toward certified passivity during aggressive contact.

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Note on verification status. Reference [14] was verified against its primary source; [1]–[12] are standard works whose bibliographic details should be checked before submission, and [13] is the under-review companion. Clinical figures (perforation threshold, cardiac-motion amplitude, tissue stiffness, sensing rates) and the Table II literature characterizations are representative and not independently verified here. Simulation results are produced by `simulation/catheter_benchmark.py` (Sections VI-B/E) and `simulation/catheter_verify.py` (Section VI-F), both single-DOF, NumPy/SciPy/OSQP.